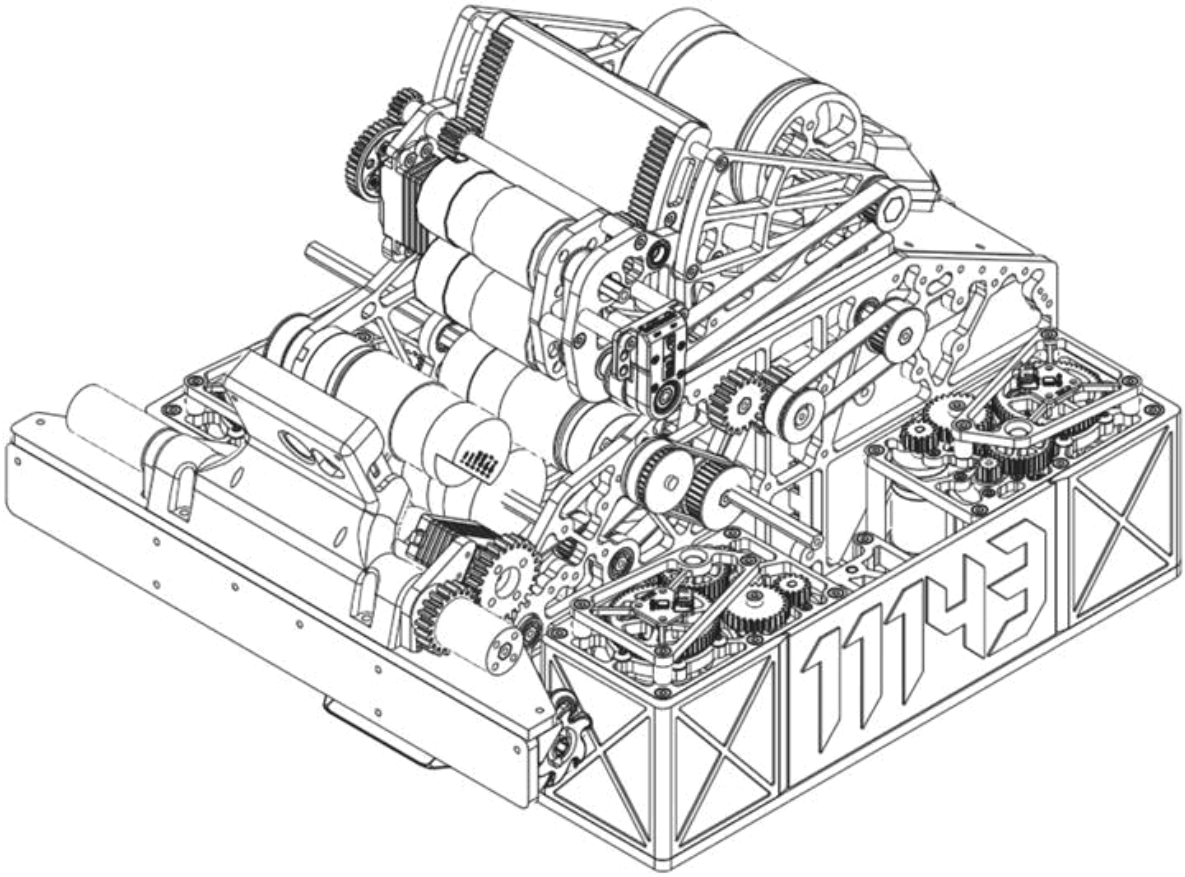
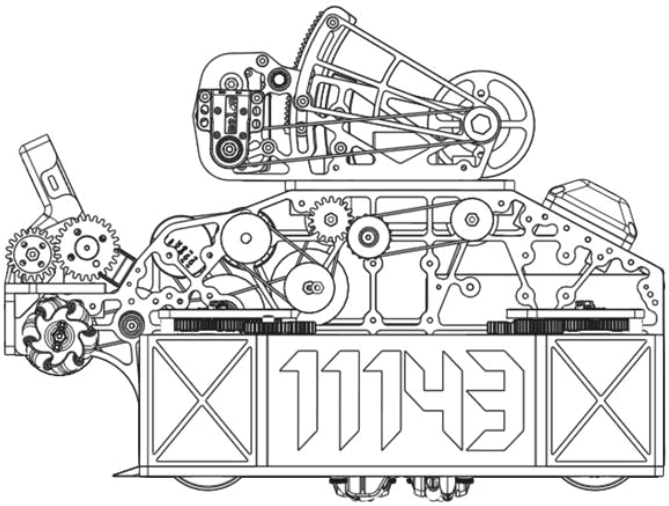
ENGINEERING PORTFOLIO

FTC - FIRST TECH CHALLENGE

BARKER YELLOWBACKS



11143



2026-2027

Sydney
Australia

Barker College
Supported by Barker Robotics

MEET THE TEAM

11143

12 MEMBERS

EB Ethan B. **LEAD DESIGN** assembly	**BL** Brandon L. **CAD** assembly	**IH** Isaac H. **LEAD CODE** CAD, assembly
RT Ryan T. **DRIVER** CAD	**WQ** Will Q. **LEAD STRATEGY** drive coach	**IW** Ivy W. **PROGRAMMING** HR
TX Tina X. **PROGRAMMING** assembly, strategy	**CW** Chelsey W. **ASSEMBLY** scouting	**CL** Chloe L. **ELECTRICAL** assembly
CS Chloe S. **STRATEGY** code, assembly	**JC** Jemima C. **PROGRAMMING** scouting	**LT** Lucas T. **CAD** assembly

HOW WE SPLIT THE WORK

- **CAD:** four of us worked on the CAD, mostly at the start of the season. Once the design was done we joined assembly so the build went faster
- **Code:** five of us wrote the basic structure of the code and our simulator. We were ready to test as soon as the robot was finished, and we kept the work moving steadily
- **Strategy:** three of us planned our strategy. We worked together on autonomous routes that suit our robot and the alliances we are likely to play with. Four of us scout
- **Assembly:** eight of us assembled the manufactured parts with the design team once the CAD was done

HOW WE WORK

- We meet **every week**
- **Subteam meetings:** each subteam shares updates from its areas. The programmers cover the shooter, the swerve and the ball transfer in turn
- **Whole-team meetings:** we share the changes and progress that affect everyone. When the design team removed the turret, we had to tell the programmers
- Whole-team meetings also keep us on schedule so deadlines do not slip
- Each subteam decides its own area, then tells everyone else

2026-2027 SEASON TIMELINE

15 SEP — Schedule
everyone adds free times

18 SEP — First meeting
subteams chosen

10 OCT — Refocus
turret cut

26 OCT-1 NOV — Swerve built
goBILDA parts arrive

12 NOV — Wollongong
first event

28 NOV — Macquarie
second event

4-5 DEC — Macquarie
national competition

28 APR-1 MAY — Worlds
FTC World Championship

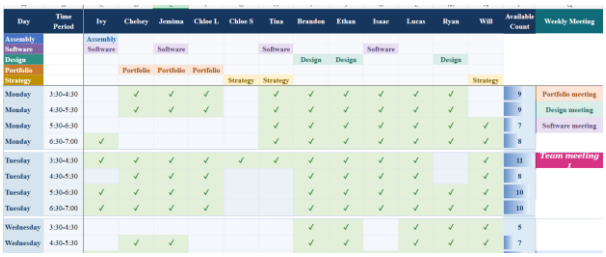
Dotted line: events after Wollongong that we are planning for.

PLANNED ON 10 OCTOBER VS WHAT HAPPENED

Our first team meeting was on 18 September. We chose our subteams and roles and began designing and programming. On 10 October we held a major meeting to refocus our timeline so the robot would be finished in time. We cut the turret (page 9) and planned each week:

WEEK	PLANNED	WHAT HAPPENED
12-18 Oct	finish the CAD; make the custom swerve parts	goBILDA parts ordered; shooter redesigned and simplified; intake and transfer design done; ball tracking for autos started; hood and shooter code started
19-25 Oct	goBILDA parts may arrive; make the shooter and transition; assembly	parts machined; assembly began
26 Oct-1 Nov	wiring, programming, fixing mechanical issues	goBILDA parts arrived; swerve assembled
2-8 Nov	programming and drive practice	<what happened>

HOW WE SCHEDULE



Our availability spreadsheet (part)

A shared spreadsheet shows who is free each week and what work they are given. It mostly worked, but an exam block and outreach made it hard to judge what each week could really fit.

OUR SEASON GOALS

- Win the national competition
- Qualify for the FTC World Championship
- <other season goals>

OUTREACH

6	Fortnightly	1,500+	32
of us teach Darkinjung students	since February 2026	people at Round Square	countries at Round Square

TEACHING DARKINJUNG STUDENTS

WHAT

Six of us (Ethan B., Brandon L., Chloe S., Ivy W., Ryan T. and Will Q.) teach robotics to Darkinjung students **once a fortnight**, since February 2026.

WHAT THEY LEARN

They built robots for this year's **VEX IQ** game and learnt how to build, how to code, how to do interviews and how to write portfolios.

IMPACT

Some of the students were in competitive teams and won **Skills**, **Teamwork Champion** and **Excellence** awards.

ROUND SQUARE INTERNATIONAL CONFERENCE

WHAT

In the Term 2-3 holidays eight of us (Ivy W., Ethan B., Ryan T., Chelsey W., Chloe L., Isaac H., Tina X. and Will Q.) ran the robotics part of the conference at Barker.

DEMOS

At the opening fair we showed robots driving, shooting, intaking and scoring.

REACH

Over 1,500 people from **32 countries** came. Many of them had not heard of FIRST before.

FOR US

It was a very rewarding experience, and it was great to see so many people there.



Our robotics showcase at Round Square

MENTORING A VEX TEAM

Three of us (Ethan B., Isaac H. and Brandon L.) mentor the Darkinjung VEX team <team name>. It is very rewarding to watch the students flourish in their games and competitions.

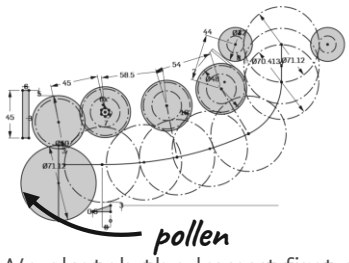
BARKER VEX COMPETITION

On **14 November 2026** Ethan B., Isaac H. and Brandon L. will referee, queue teams and help the competition run smoothly.

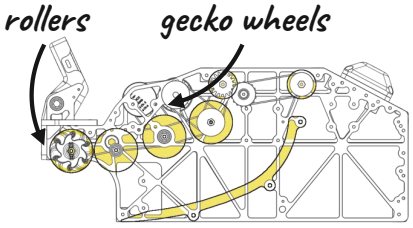
OUR DESIGN LOOP

1. **Goals** — what the game needs
2. **Sketch + CAD** — Onshape layouts
3. **Simulate** — same code, desktop sim
4. **Build** — parts and assembly
5. **Test + change** — back to step 2

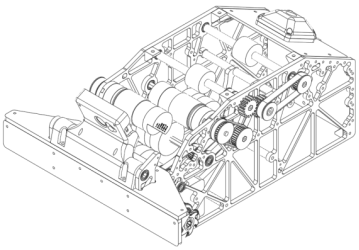
OUR CAD PROCESS, SHOWN ON THE TRANSITION



1. We sketch the layout first, with the pollen and roller sizes



2. We design the plates and belts around the sketch



3. We assemble the whole transition in CAD

CODING TOGETHER

We each code one subsystem on its own branch, so we do not overwrite each other's work. Tina X. wrote the intake and Jemima C. wrote the shooter. On **25 October** we merged everything into one robot, which showed us some problems:

- Two branches each hid the other's subsystem
- We were mixing units (RPM, ticks, degrees), so every value now has one unit
- The other fixes are on page 12

IDEAS FROM VEX AND LAST SEASON

Four of us (Ethan B., Will Q., Ryan T. and Tina X.) competed in VEX **Push Back**. The match loaders in that game worked well for getting pollen out of the flower, so we used the same idea (page 10).

Swerve and logging carried over from last season.

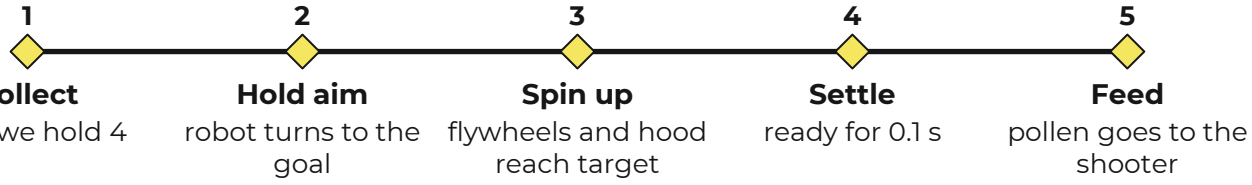
WHAT SCORES POINTS (MANUAL, TABLE 10-2)

SCORING ACTION	POINTS	OUR ANSWER
HIVE TIP, by launching into the upward cell	20 in auto and 20 in teleop	shooter, hood and auto-aim
Pollen left in an upward cell at the end	2 each	the same launches

HOW WE PLAY

<our scoring plan: what we score first, where we shoot from, our role with the alliance partner>

ONE SCORING CYCLE



REQUIREMENTS, AND HOW THE ROBOT MEETS EACH

WE NEED TO	WHY	OUR ANSWER
Launch from anywhere	fewer trips across the field	moving hood
Launch quickly and often	less waiting between shots	two flywheels
Launch without turning round	pollen collects along the wall	rear-facing shooter
Aim while driving	save time every cycle	swerve and auto-aim
Hold our spot	defence will push us	swerve X-lock
Avoid defence	defenders will chase us	very small frame perimeter
Stop opponents shooting	fewer points for them	blocker
Empty the flower	pollen sits inside it	flower mechanism
Find pollen and the goal	drive and aim without guessing	front and rear cameras

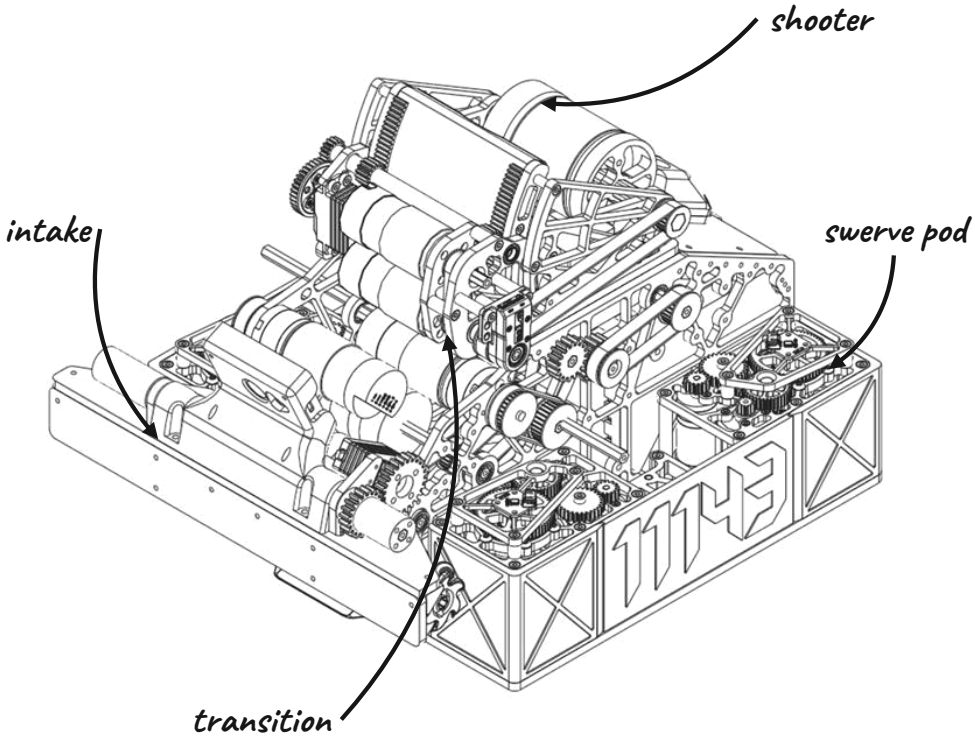
How well it worked

<a few sentences on how well each answer worked on the robot>

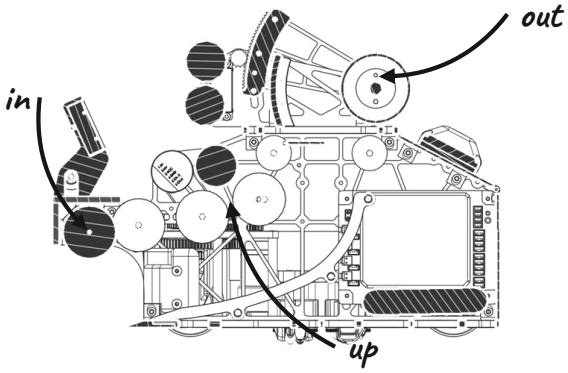
SCOUTING

Jemima C., Chelsey W., Chloe L. and Tina X. scout. <what we record and how it changes our plan>

OUR ROBOT



POLLEN PATH

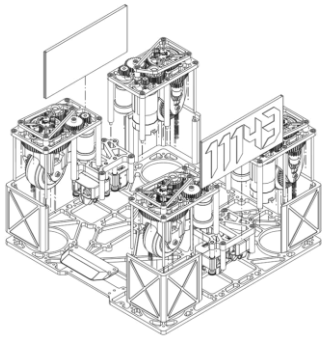


In at the front, up the transition, out of the rear shooter

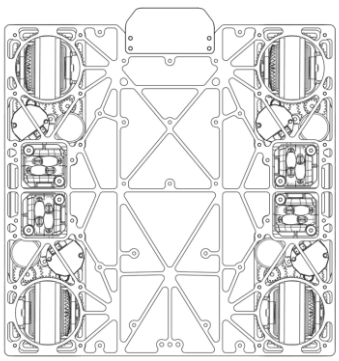
DESIGN CHOICES

- A **very small frame perimeter**, so defence is easier to avoid
- **Swerve drive:** we move in any direction and can lock the wheels in an X
- A **rear-facing shooter**, so we can launch straight after collecting along the wall
- A **light robot**, so it moves more efficiently

CHASSIS



Exploded view: pods, plates and side panels

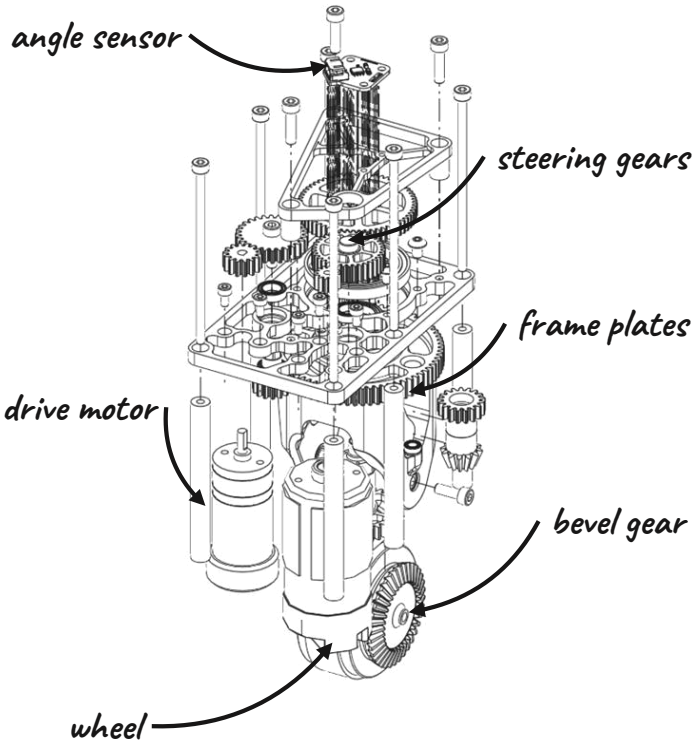


Underneath: a pod in each corner

SWERVE DRIVETRAIN

11143

ONE SWERVE POD



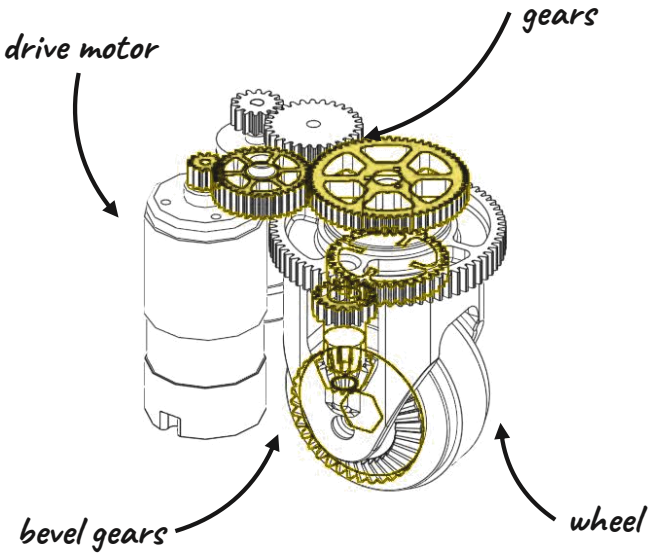
WHY SWERVE

- **X-lock:** we can place the wheels in an X. The friction makes us much harder to push, which helps against defence
- We move more efficiently, which is a big advantage in a match
- Full-grip wheels give more speed and pushing power than mecanum
- We can drive in one direction while facing the goal
- We built swerve last season, so we already knew it

Our pods

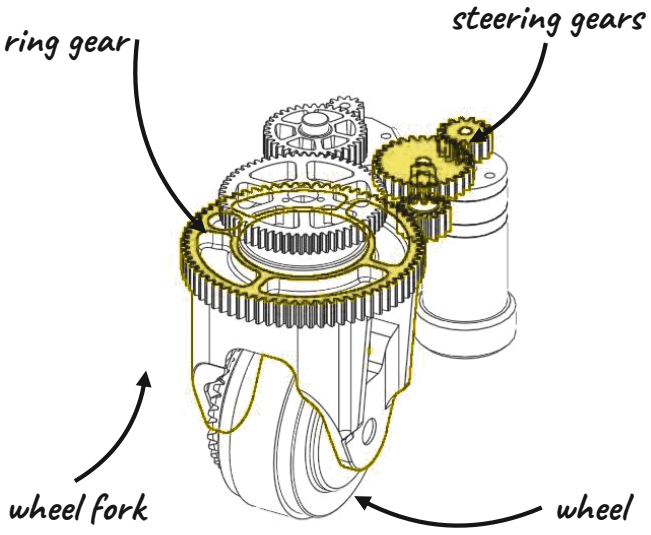
- Each pod has a drive motor and a steering servo
- We designed and machined custom parts for the pods
- Every pod has an **absolute angle sensor**, so each wheel knows where it points from the start

WHEEL DRIVE



Motor, gears, then a bevel gear turn the wheel

MODULE ROTATION

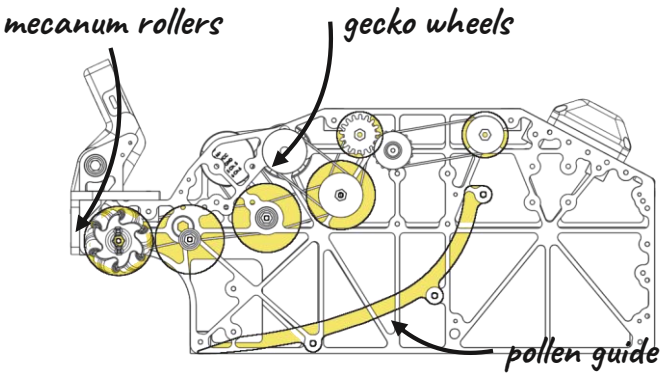


A servo and a ring gear turn the whole pod

TESTS

TEST	RESULT	STATUS
Top speed	not finished yet	**NOT TESTED**
Pushed by another robot in X-lock	not finished yet	**NOT TESTED**

POLLEN PATH

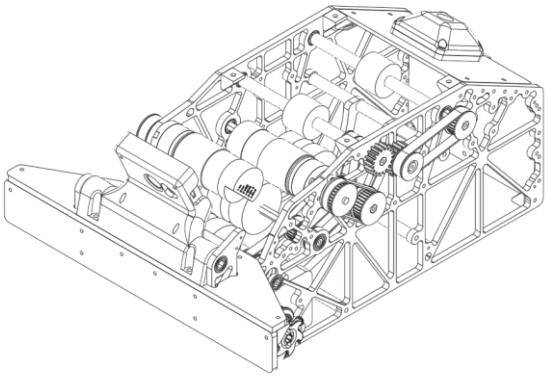


Rollers pull pollen in and gecko wheels lift it to the shooter

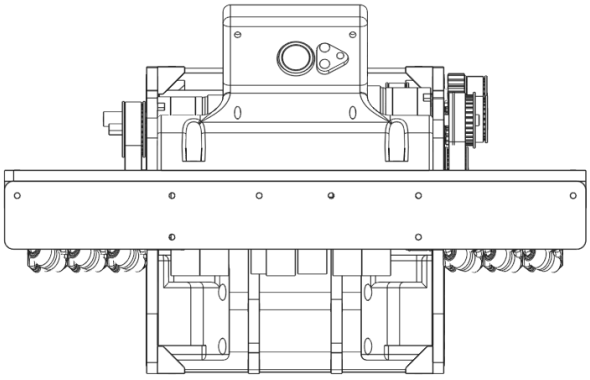
HOW IT WORKS

- Mecanum rollers at the front pull pollen in
- Gecko wheels in the transition grip the pollen and carry it up
- Smaller gecko wheels at the top reduce **backspin** going into the shooter
- **Polycarbonate ramps** give the pollen a low-friction surface
- Two motors drive the intake and the transition
- **4 beam sensors** count the pollen we hold, up to the 4 that rule G407 allows. The intake stops when we are full

ASSEMBLY



The intake and transition together



From the front

TESTS

TEST	RESULT	STATUS
Jams per 20 pickups	not finished yet	**NOT TESTED**

SHOOTER DESIGN

WHAT WE TRIED

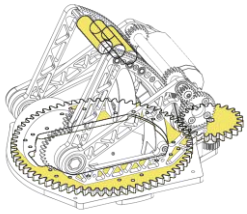
TURRET, LAST SEASON

+ aims without turning the robot

− months of tuning (Apr-Jun 2026)

→

TURRET, THIS SEASON



+ what we wanted first

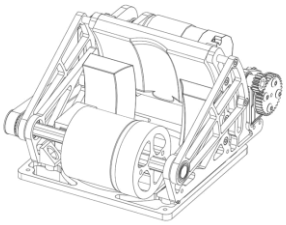
− weight and space

− goBILDA parts delayed

− no time: regionals 4 weeks away

→

FIXED REAR SHOOTER



+ fewer parts

+ shoots out of the flowers without turning

− the whole robot must turn to aim

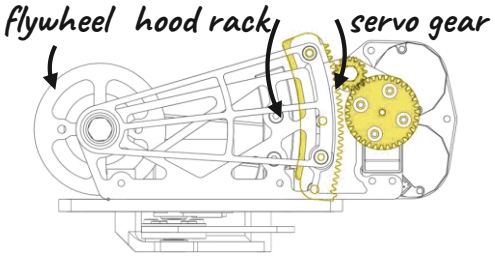
WHAT WE WANTED, AND WHY

- **Two flywheels,** so there is less waiting between shots
- **A moving hood,** so we can shoot from anywhere on the field. It moves from 33.6 to 67 degrees
- **A shooter that faces backwards.** Pollen collects along the wall, so we can drive up to the wall and aim without turning round first
- **A lookup table** that gives the flywheel speed and hood angle for where localisation says we are. We will measure its values once the robot is finished

Why we cut the turret

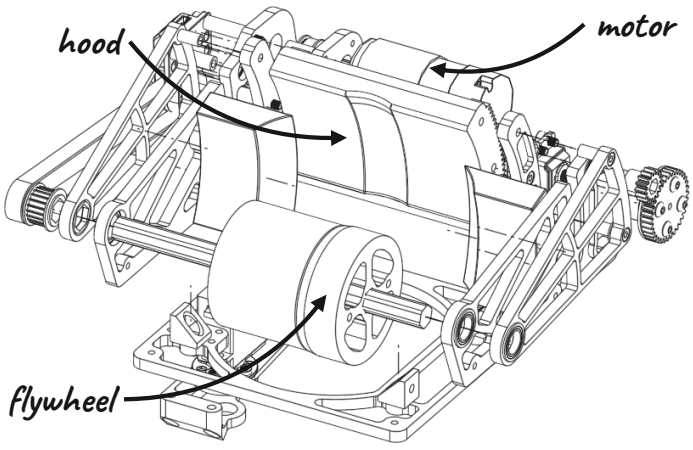
At our meeting on 10 October we decided to cut the turret. Regionals were four weeks away and our goBILDA parts were delayed, so we needed to save time. The turret was the least necessary part. We kept swerve and the blocker, which still speed up shooting, and spent the time making the intake, transfer, swerve, autos and code consistent. The swerve base now turns the robot to aim (page 13).

HOOD DRIVE



A servo gear drives a rack to tilt the hood

EXPLODED VIEW



TESTS

TEST	RESULT	STATUS
Shots in from 3 distances	not finished yet	NOT TESTED
Time between shots	not finished yet	NOT TESTED

FLOWER, BLOCKER, CAMERAS

FLOWER MECHANISM

IDEA

Four of us competed in VEX **Push Back**. The match loaders in that game worked well for getting pollen out of a container, so we used the same idea for the flower.

PROBLEM

Pollen sits inside the flower and we need to get it out.

RESULT

Not tested yet: the robot is not finished.

BLOCKER AND GATE

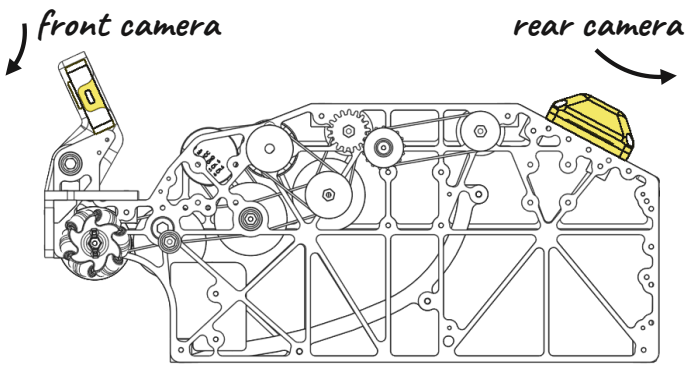
BLOCKER

The blocker stops opponents from shooting (1 Axon Max servo).

GATE

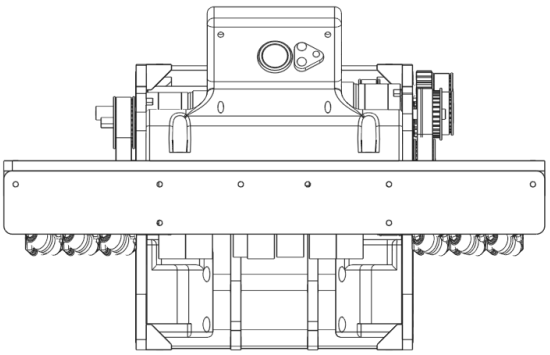
The gate stops pollen going into the shooter until the shot is ready (1 Axon servo).

TWO CAMERAS



- The **front camera** finds pollen so we can drive a path to it
- The **rear camera** reads AprilTags so we know where the goal is
- Status: not tested yet (robot not finished)

HOW WE HOLD AND RELEASE POLLEN

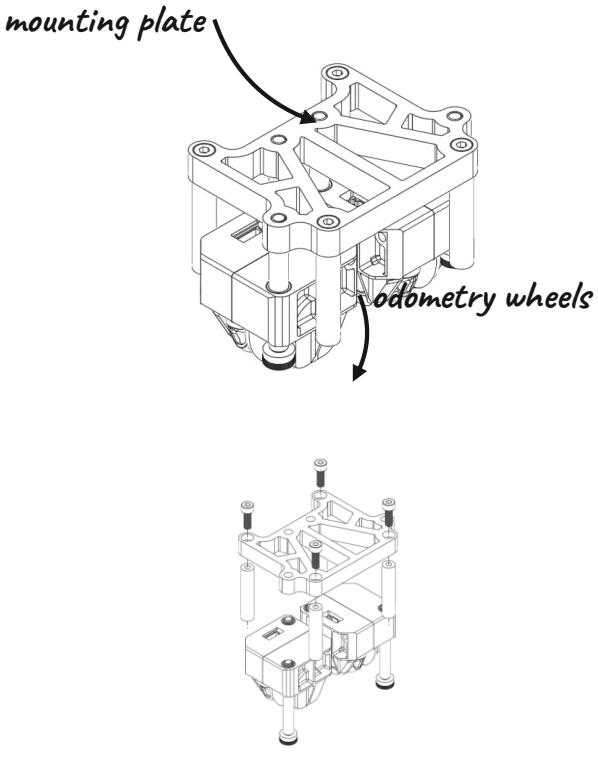


- **Collect:** the intake runs until the 4 beam sensors say we are full, then it stops
- **Hold:** pollen waits while the robot aims
- **Release:** pollen only feeds once aim, flywheel speed and hood have all been ready for **0.1 s** (page 13)

WHAT EACH PART DOES

PART	PROBLEM IT SOLVES	WHAT IT DOES
4 drive motors	moving in any direction	drive each swerve wheel
4 steering servos + 4 angle sensors	pointing every wheel the right way	turn each pod and read its angle
IMU	knowing which way we face	heading for driving and aiming
Front and rear cameras	finding pollen and the goal	pollen pathing; AprilTags
4 beam sensors	knowing how much pollen we hold	count the pollen along the transition
Gate servo	pollen reaching the shooter too early	holds pollen until the shot is ready
2 shooter motors	the same shot every time	flywheels held at target speed (PIDF)
Hood servo	shooting from different distances	sets the launch angle
Flower mechanism servo	pollen stuck inside the flower	gets it out
Blocker servo	opponents shooting at us	stops their shots
Voltage and current checks	brown-outs and stalled motors	logged every loop
8 motors and 8 servos	rule R503 limits us to 8 of each	we use all 16

ODOMETRY POD



Designed in CAD. The code uses wheels and the IMU for now

SPEED CONTROL

CHALLENGE

Each shot takes energy from the flywheels, and a slower wheel shoots short.

SOLUTION

A **PIDF** controller holds the target speed. Its feedforward uses battery voltage, so a flat battery does not change the shot.

LIMITS

The flywheels are limited to **3500 RPM**. They count as ready within **50 RPM** of the target and the hood within **2 degrees**.

RESULT

Not tested yet: the robot is not finished.

SAME CODE ON THE ROBOT AND ON A LAPTOP

PART	WHAT IT HOLDS	WHY
RobotCode	behaviour: driving, aiming, shooting	written once, used on both
TeamCode	the real motors, servos and sensors	only place that touches hardware
RobotSim	simulated robot, field and pollen	test before the robot is finished

WHY WE SIMULATE

CHALLENGE

The robot is not finished yet, but the code needs testing now.

SOLUTION

The same TeleOp runs in our simulator on the official field CAD.

RESULT

In simulation we score **3 of 3** for both alliances, also while moving.

AUTOMATED TESTS

- **24 desktop tests** pass (25 October 2026)
- They cover heading wrap, 10 ms loops, a lost heading sensor, manual turn override and stop outputs
- Every loop is logged, so we can replay a run

WHAT WE FOUND AND FIXED WHEN WE MERGED THE CODE

PROBLEM	FIX
Shooter could never reach its shooting state	a target speed now starts the wheel controller
Updates changed numbers but moved nothing	read feedback, write a limited motor power every loop
RPM, encoder ticks, degrees and servo positions mixed	one unit for each value; conversion only in hardware code
The flywheel's controller also drove the hood	the hood has its own mapping and rate limit
Lookup table divided by zero at its first point	handles exact points; never guesses past its ends
Simulated aiming stopped about 12 degrees short	wheel feedforward based on battery voltage

RISKS AND HOW WE REDUCE THEM

RISK	HOW WE REDUCE IT
Simulation is not the real robot	the code refuses to feed until the shooter is calibrated
Heading sensor lost mid-match	tested in simulation: stops driving and feeding
Flat battery changes the shot	feedforward scaled by battery voltage
A complex turret fails	no turret: the drivetrain aims

AUTO-AIM

THE PROBLEM

CHALLENGE

We dropped the turret, so the shooter faces backwards and cannot turn on its own.

SOLUTION

Holding the aim button turns the **whole robot** until it is within **2.5 degrees** of the goal.

WHAT THE ROBOT CHECKS BEFORE EVERY SHOT, IN ORDER

CHECK	IF NOT, THE ROBOT SAYS	WHY
Is there a possible shot?	NO TRAJECTORY	do not waste pollen
Is the shooter calibrated?	CALIBRATION REQUIRED	simulation is not the real robot
Is it aimed and not still turning?	AIMING	within 2.5 degrees
Flywheels and hood on target?	SPINNING UP	within 50 RPM and 2 degrees
Ready for 0.1 s?	SETTLING	it must stay on target, not pass through it
Is there pollen to shoot?	EMPTY	the beam sensors know
All yes	FEEDING	pollen goes to the shooter

MATHS: TURNING TO AIM

Turn speed = **feedback** on the heading error + **damping** + **feedforward** for a goal that moves as we drive + a small push past **static friction**.

Gains: kP 4.0, kD 0.35, kS 0.06.

In simulation, aiming first stopped **12 degrees** short because small commands sat inside the static friction dead zone.

MATHS: SHOOTING WHILE MOVING

The pollen leaves with the shooter's speed **plus the robot's**:

v pollen = v shot + v robot + turn rate × r

(r is where the pollen leaves, measured from the robot's centre). We look **0.07 s** ahead to where the robot will be at release, then a gravity solver finds the speed and hood angle.

TESTS

TEST	SIMULATION	ROBOT
Score while still	3 of 3	not finished yet
Score while moving	3 of 3	not finished yet

OUR ROUTE

<auto route drawing: our hand-drawn route, redrawn cleanly>

EACH STEP

STEP	WHAT	POINTS
1	<step>	<pts>
2	<step>	<pts>
3	<step>	<pts>

STATUS

- On 12-18 October we started a **ball-tracking** setup so the robot can find, map and intake pollen by itself
- The full autonomous is **not finished**: NOT TESTED
- <date we plan to have it running>

WHAT IT CAN USE

PART	WHAT IT GIVES THE AUTONOMOUS
Wheels + IMU	where we are as we drive
Rear camera	AprilTags, to know where we are on the field
Front camera	a path to the pollen
Beam sensors	how much pollen we hold
Auto-aim	the same shot checks as TeleOp (page 13)

TESTING AND LESSONS

TEST RESULTS

TEST	RESULT	STATUS
24 automated code tests	all pass	**PASS**
Simulation: score 3 of 3, still and moving	3 of 3	**PASS**
Simulation: pick up 3, aim, score 3	3 of 3	**PASS**
Simulation: lost heading sensor	stops drive and feeding	**PASS**
Robot: shots in from 3 distances	not finished yet	**NOT TESTED**
Robot: time between shots	not finished yet	**NOT TESTED**
Robot: top speed	not finished yet	**NOT TESTED**
Robot: X-lock push	not finished yet	**NOT TESTED**
Robot: jams per 20 pickups	not finished yet	**NOT TESTED**
Autonomous	not written yet	**NOT TESTED**

LESSONS WE APPLIED

- A turret cost us months last season, so this season the drivetrain aims
- We sketch the pollen path before the CAD, starting with sizes
- Every value in the code has one unit (RPM, radians)
- We merge subsystems early, because the bugs on page 12 only showed when they met
- When time is short, we cut the least necessary part and make the rest consistent
- We plan each week around exams and outreach

NEXT

- Finish the robot, then run every test on this page
- Calibrate the shooter on the real robot
- Measure the lookup table
- Write and test the autonomous
- <next design change>

RELIABILITY

No matches yet: the robot is not finished. For every match we will record whether it worked all match, and any fault and its fix.

Endnote: we drafted and laid out this portfolio with help from an AI assistant. All results, photos, CAD, code and decisions are our own, and we checked every page.